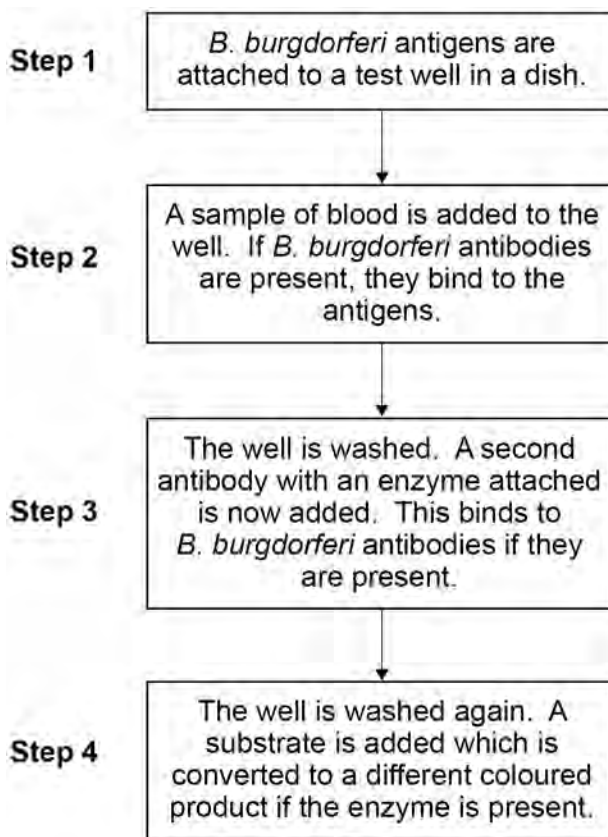


0 8

Lyme disease is most frequently caused by the bacterium *Borrelia burgdorferi*. Lyme disease can be difficult to diagnose.

Figure 5 shows an ELISA test that is used to find out if a person has antibodies to *B. burgdorferi*.

Figure 5



A false positive in this test is a result which incorrectly indicates that antibodies to *B. burgdorferi* are present.

0 8 . 1

Failure to thoroughly wash the well in **Step 4** can result in a false positive.

Explain why.

[2 marks]

Turn over ►



08.2

A false positive can be produced if a person has been infected by another bacterium that causes a disease called syphilis.

Suggest why.

[1 mark]

08.3

A false **negative** in this test is often produced if a person is tested within 2 weeks of being infected with *B. burgdorferi*.

Explain why.

[2 marks]

08.4

Sometimes, symptoms of Lyme disease can persist for 6 months following antibiotic treatment. This condition is known as Post-Treatment Lyme Disease Syndrome (PTLDS).

Scientists investigated the symptoms experienced by a large number of PTLDS patients and a control group. During a 2-week period, they asked all the participants:

- if they had experienced symptoms of PTLDS
- to record the intensity of these symptoms.

The scientists used a statistical test to determine if there was a difference in the intensity of symptoms of PTLDS between these two groups.



